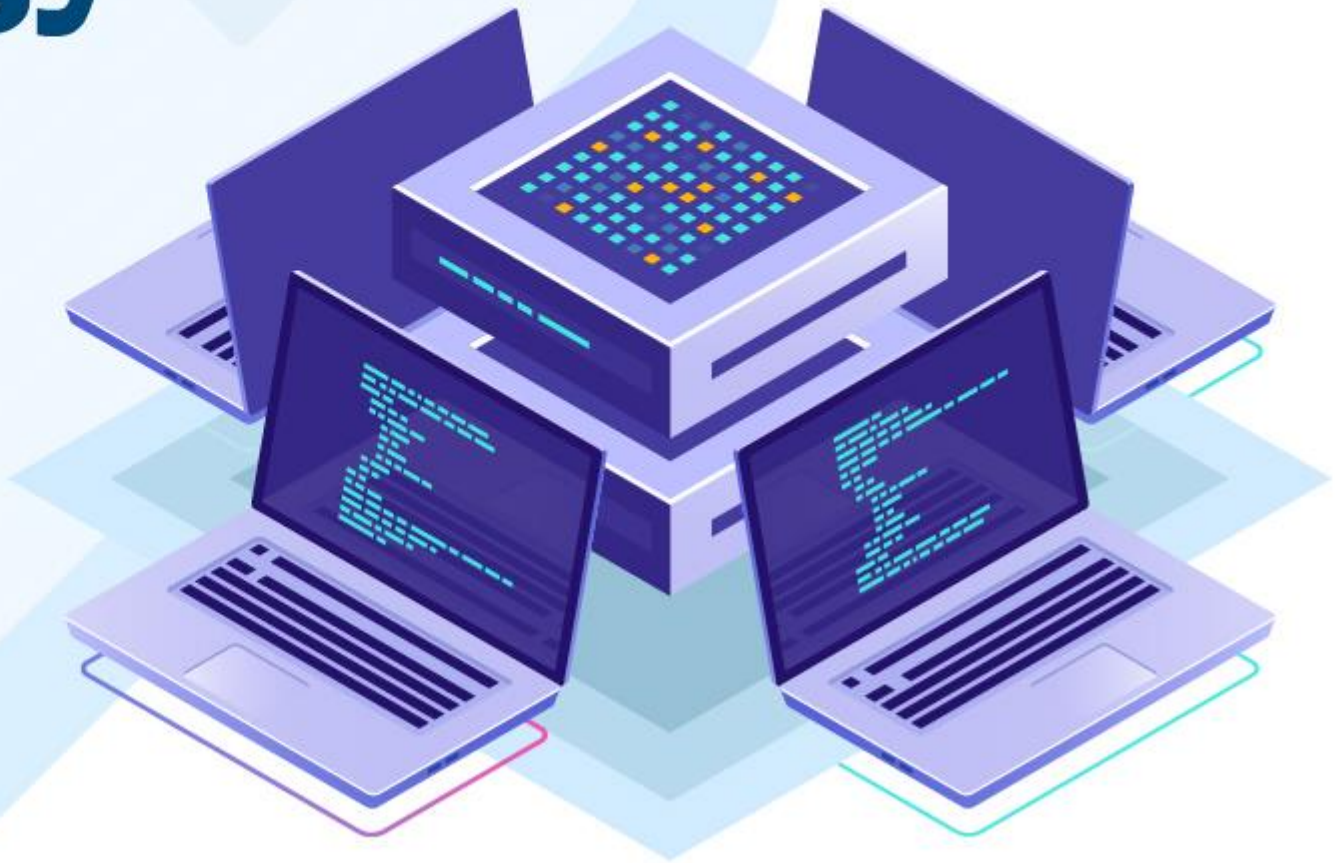
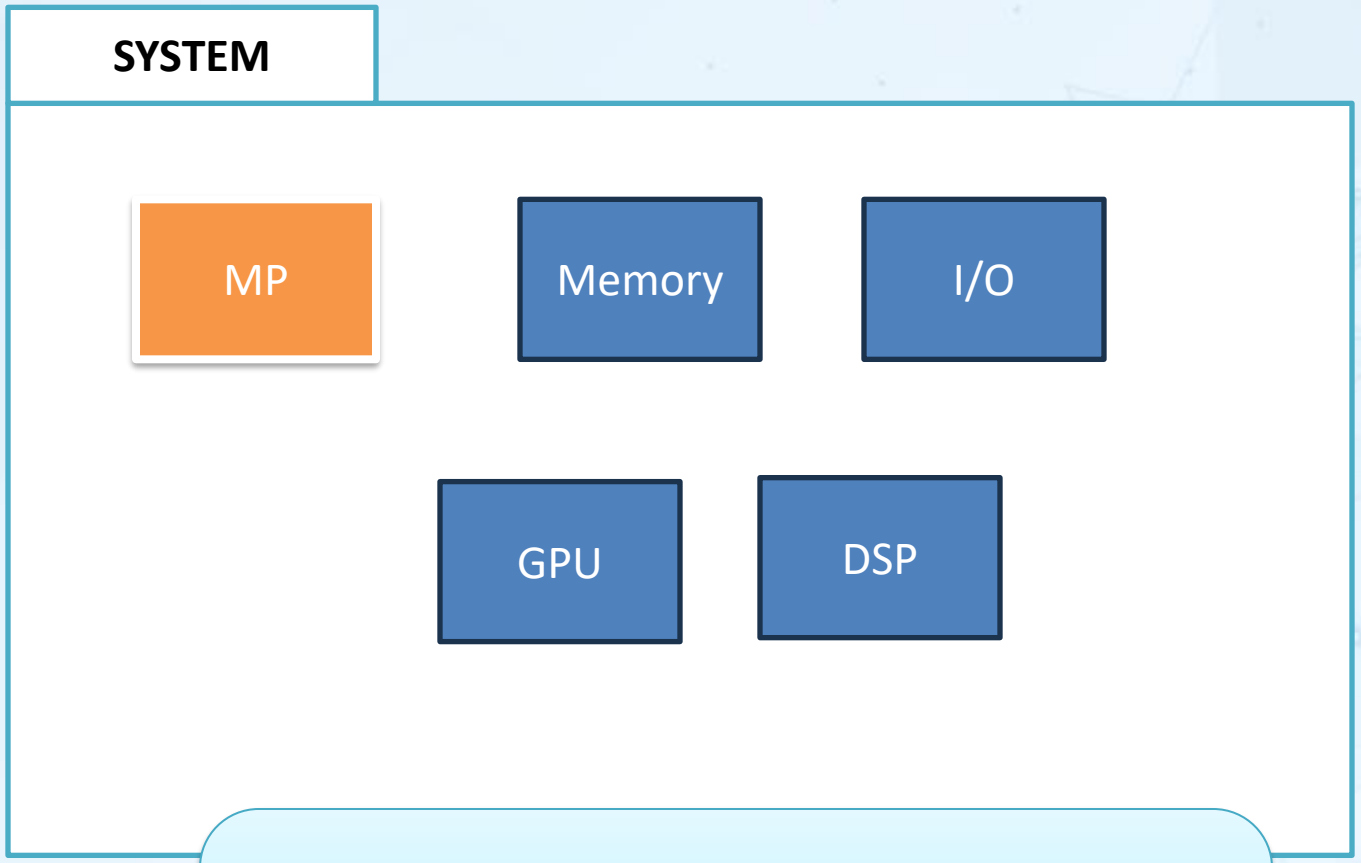
Information Technology

Fall Semester
2026 - 2027

- Embedded Systems
- **Dr. Yasmine Mahmoud**

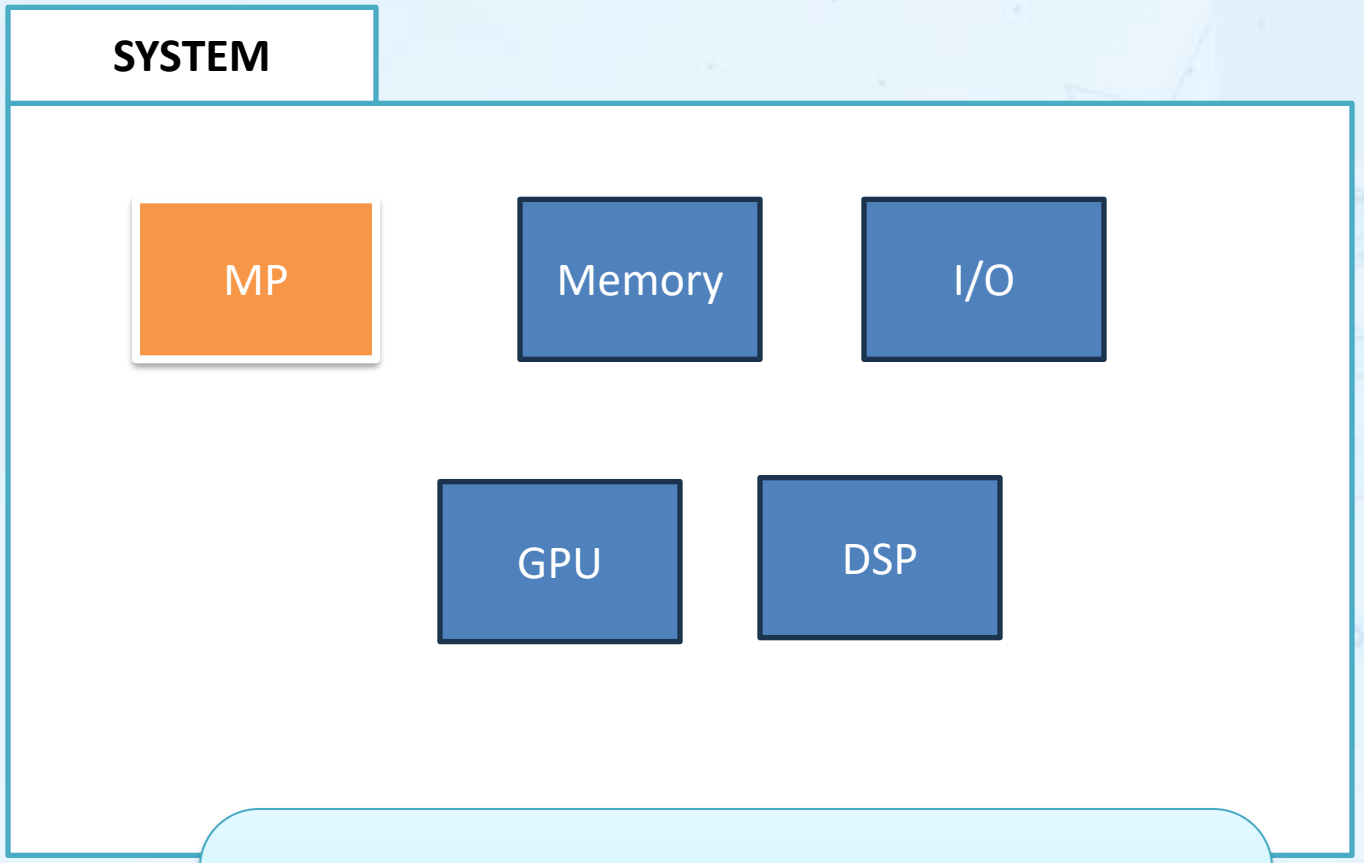


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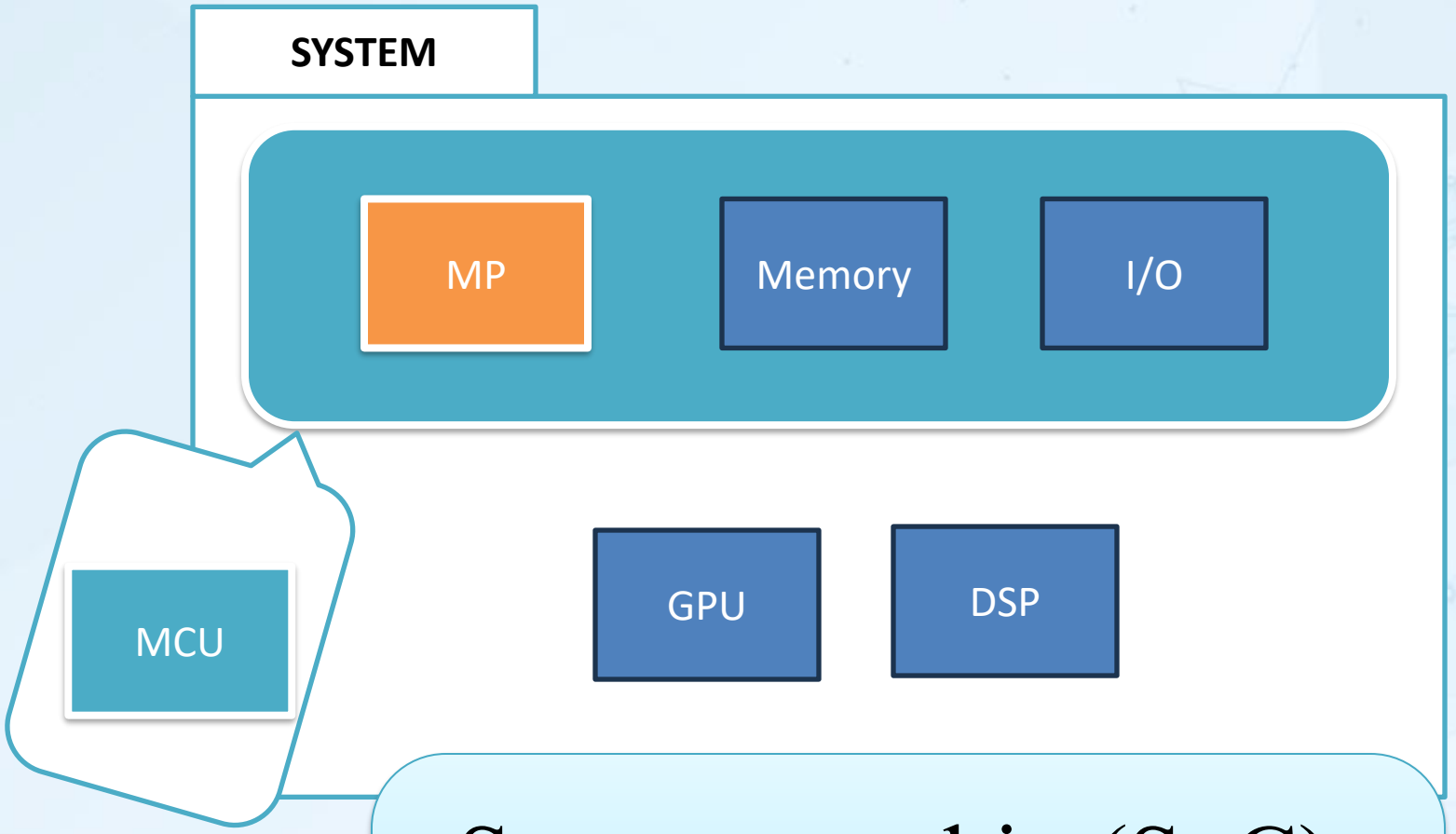
Name this system?

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System on chip (SoC)

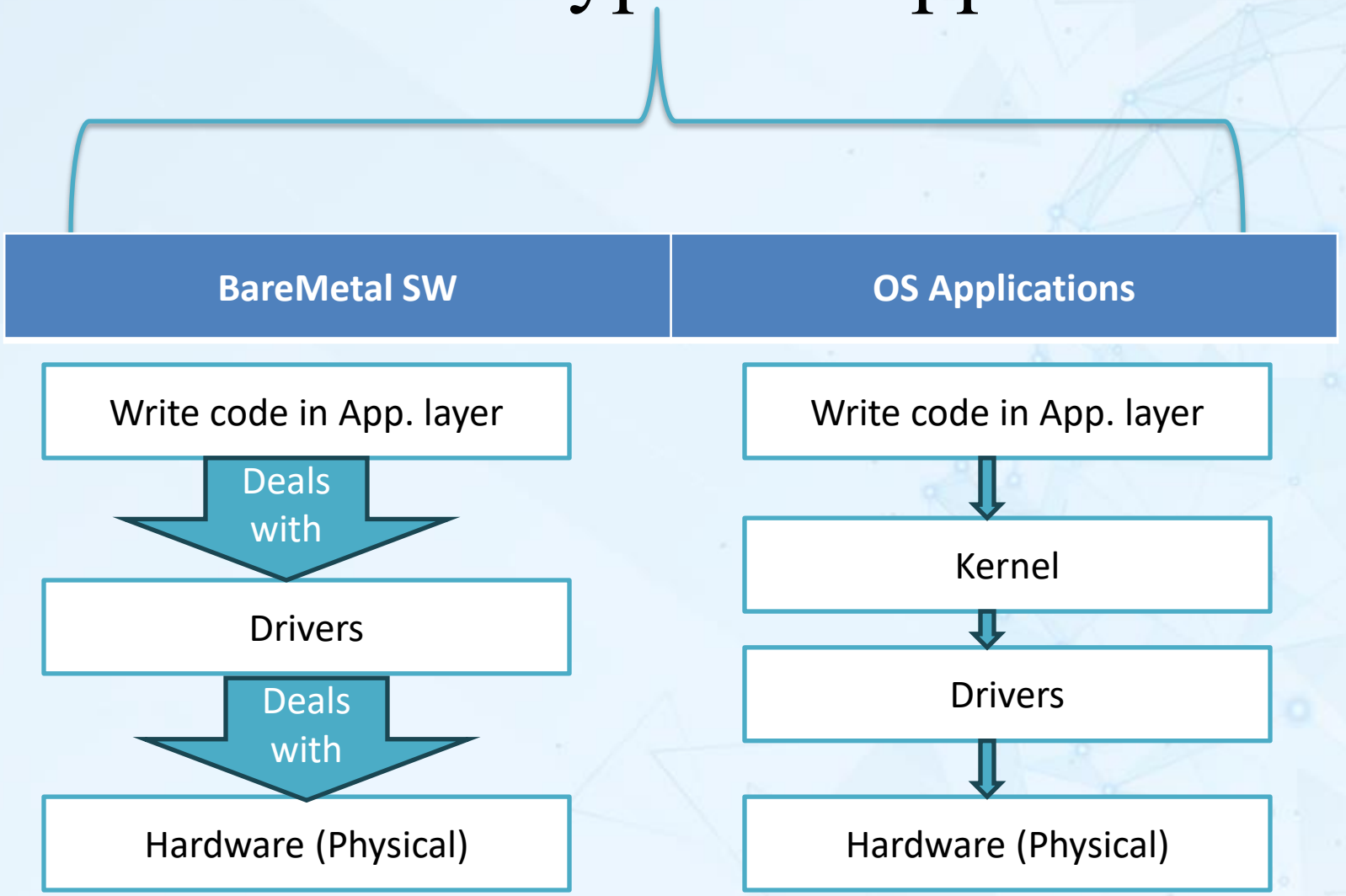
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System on chip (SoC): is a high performance MCU

Applications

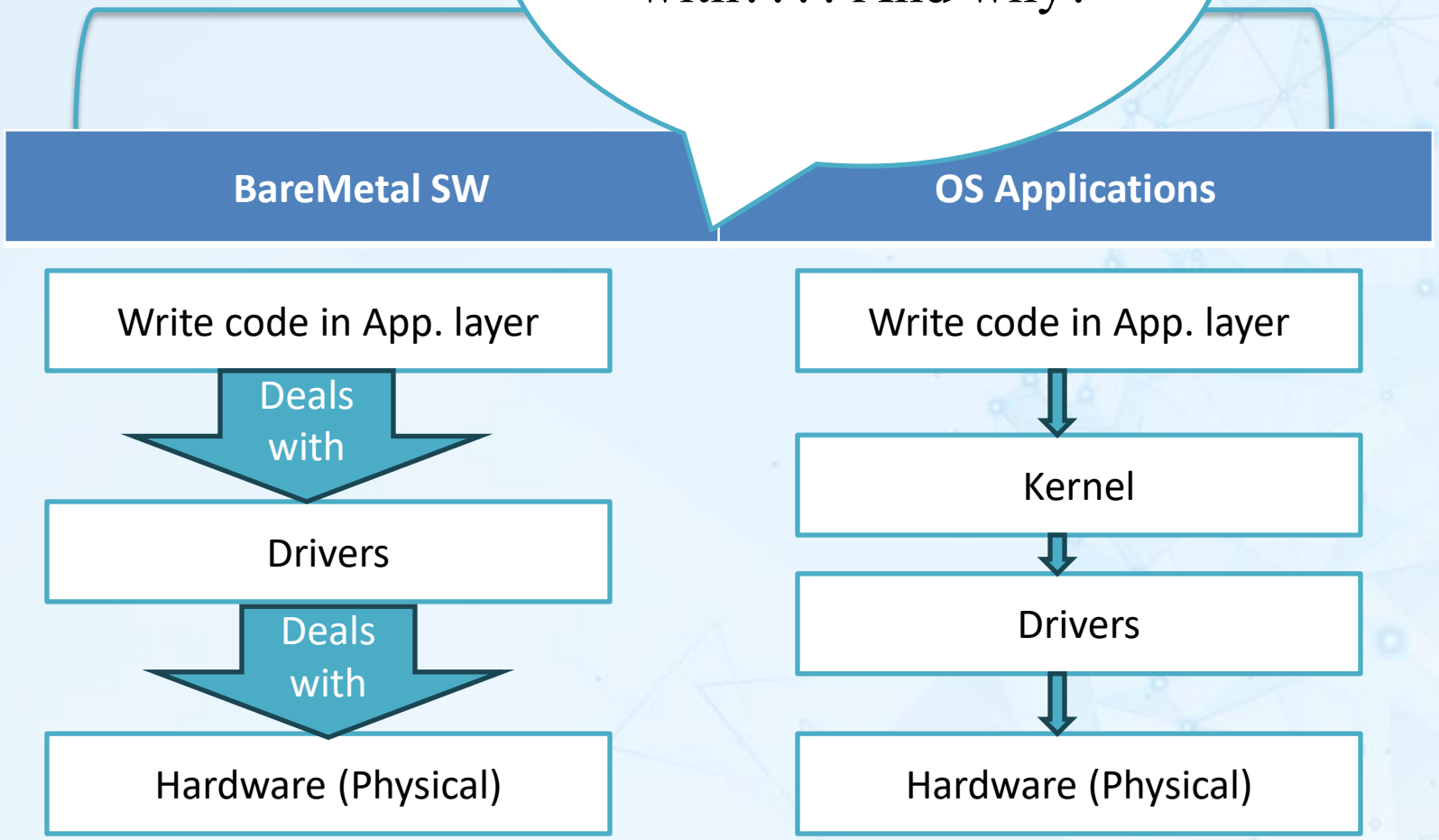
There are two types of Applications:



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A

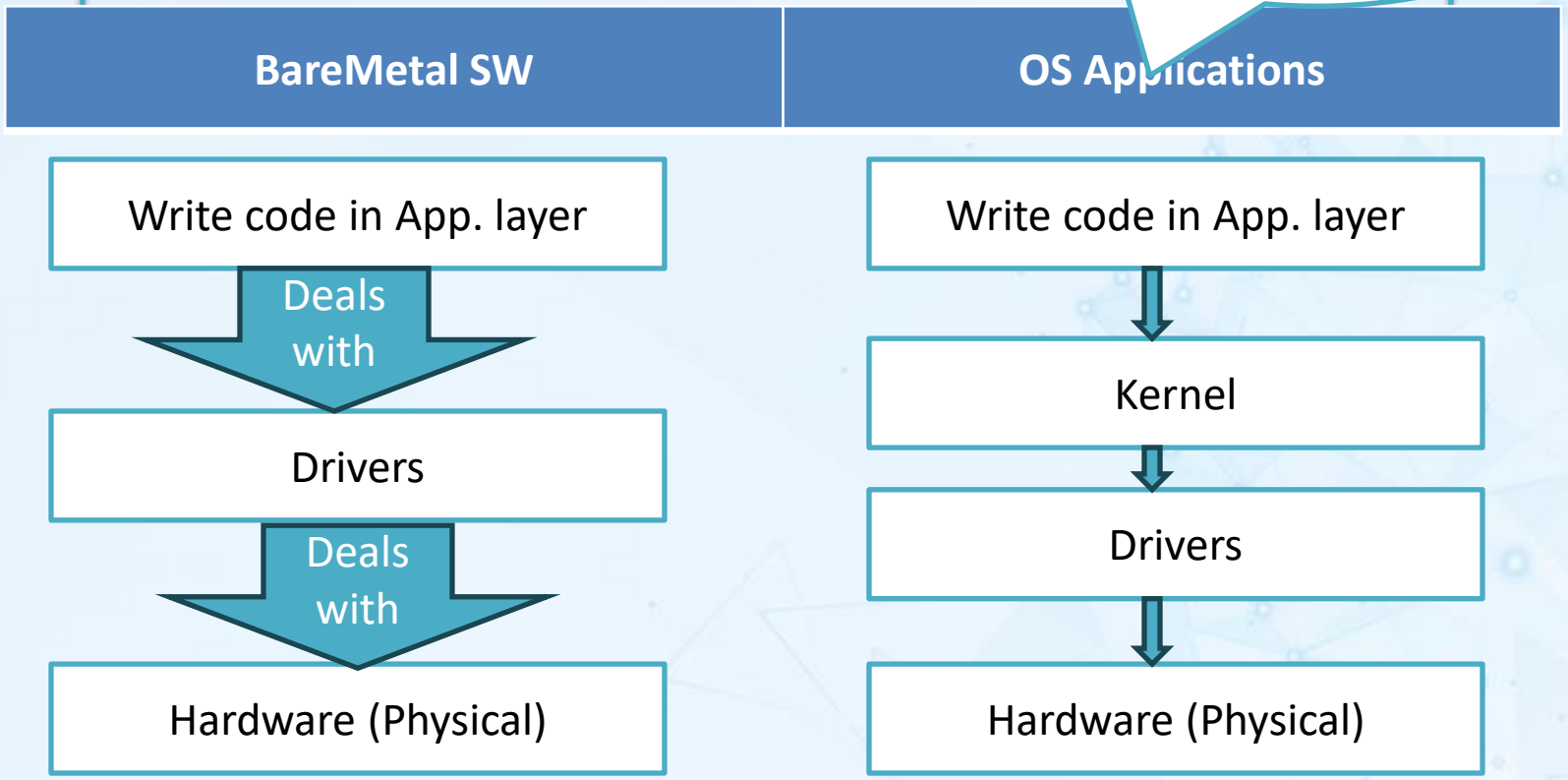
Which one System on Chip (SoC) deals with???? And why?



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Applicati

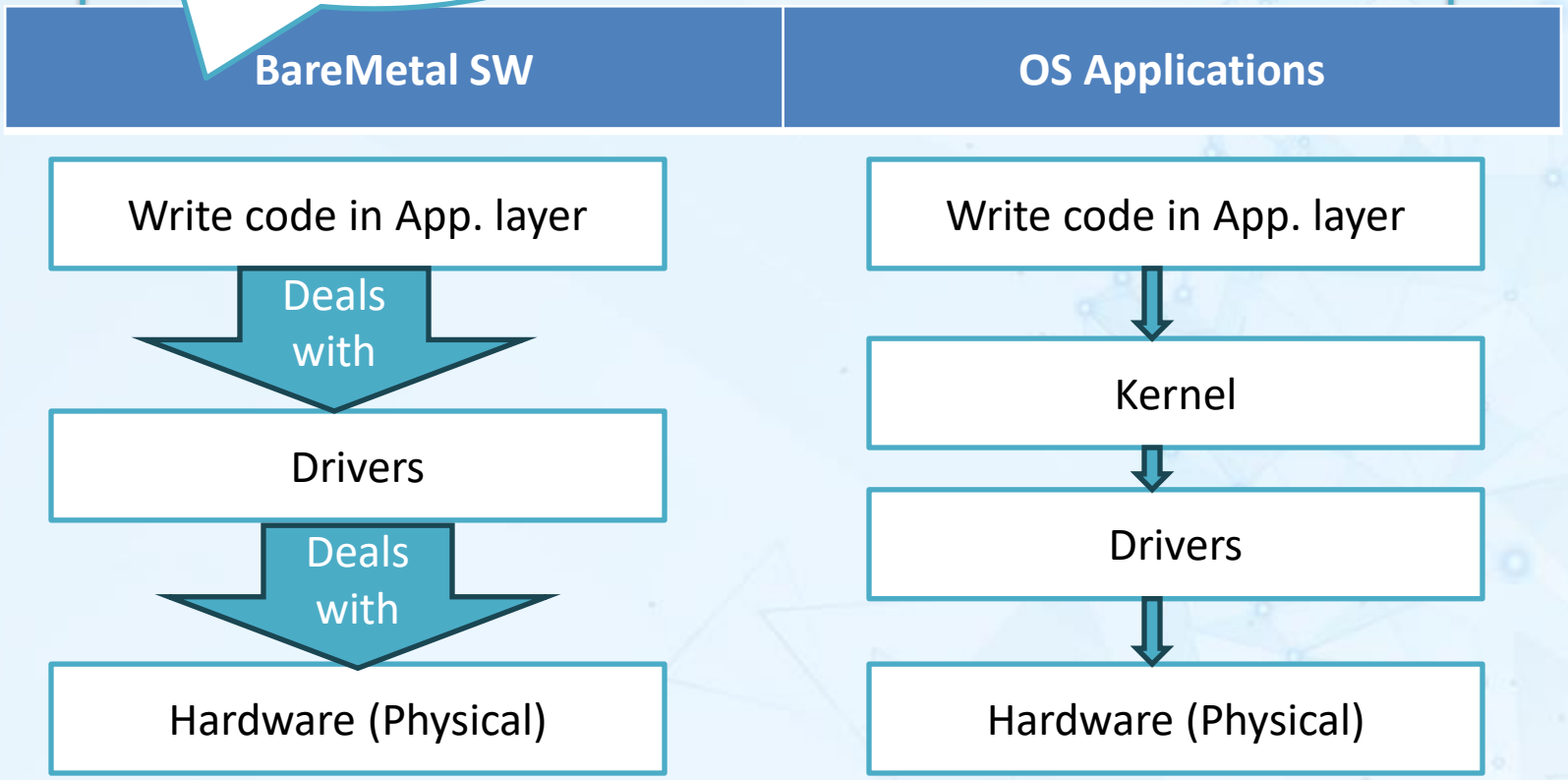
SoC deals with OS Applications , because we need Microcontroller with high performance



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ications

We will deal with BareMetal SW, as we deal with **not** highly performance microcontroller like SoC



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Applications

OS Applications

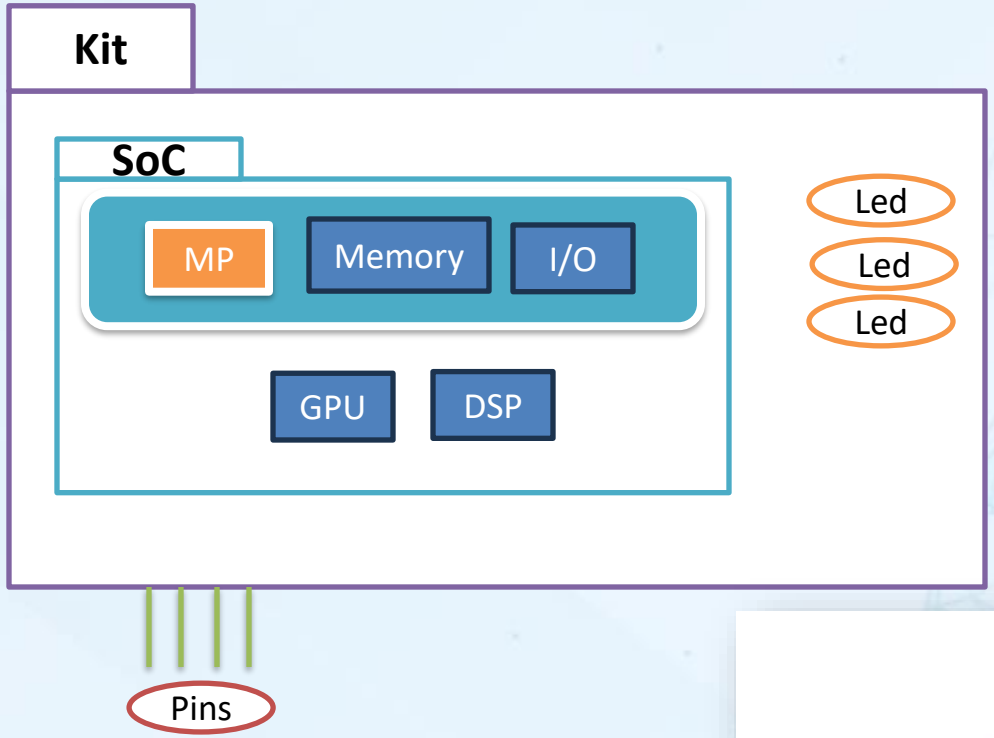
Non-BareMetal SW
Ex.
- Embedded Linux
- Android

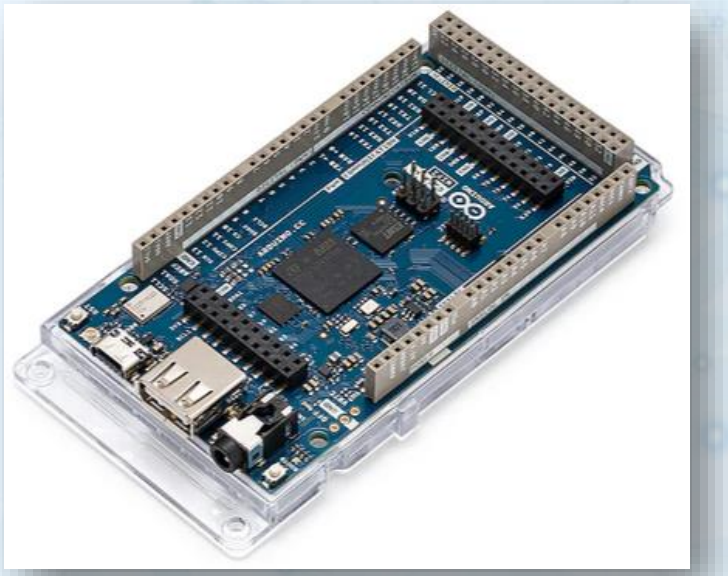
(Deals with SoC)

Kit

- Ex. Arduino kit



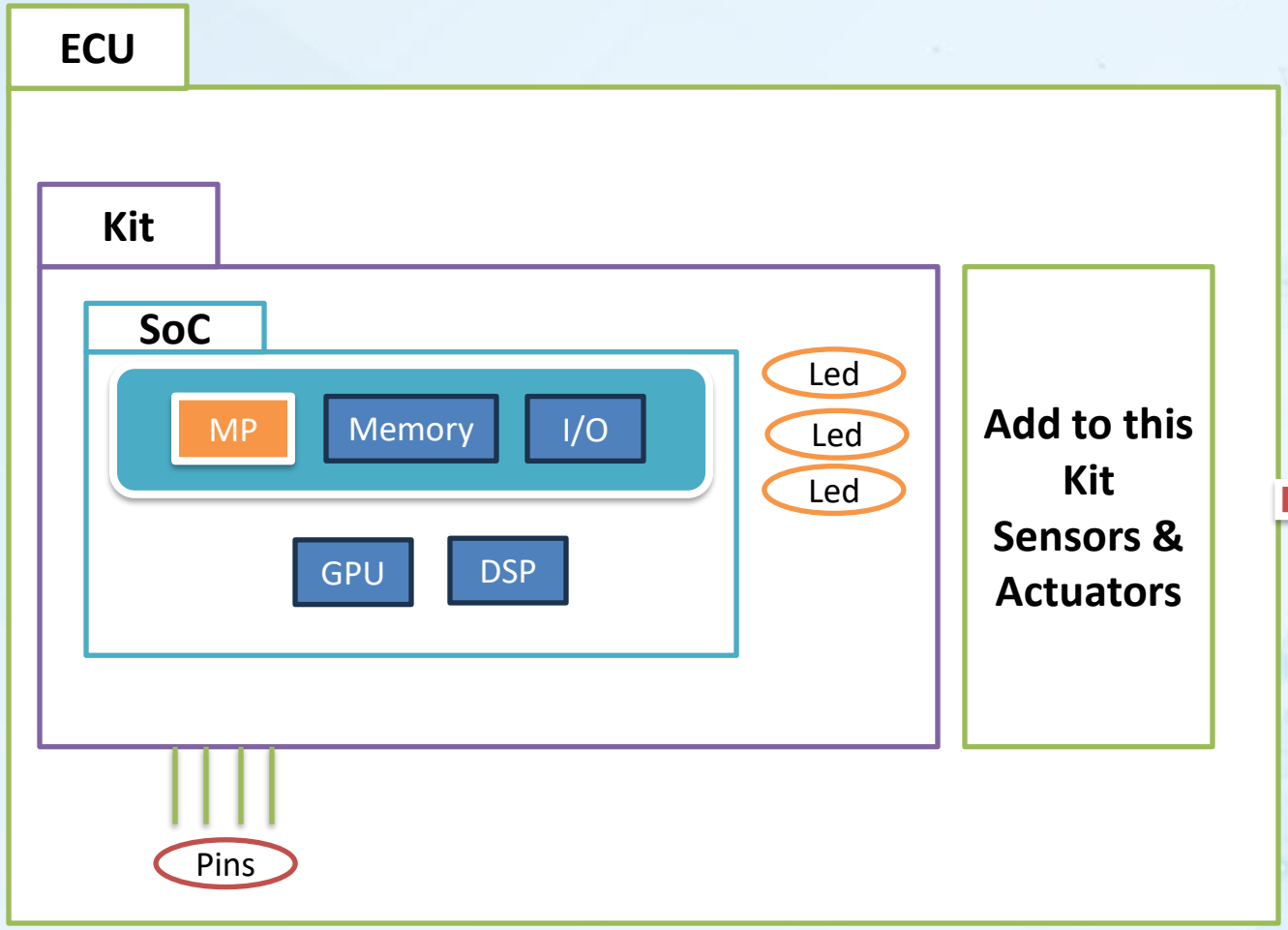




- Ex. Arduino kit

(Microcontroller on Kit + bins)

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Electrical Control Unit

Large systems like cars, consists of at least 200 ECU.

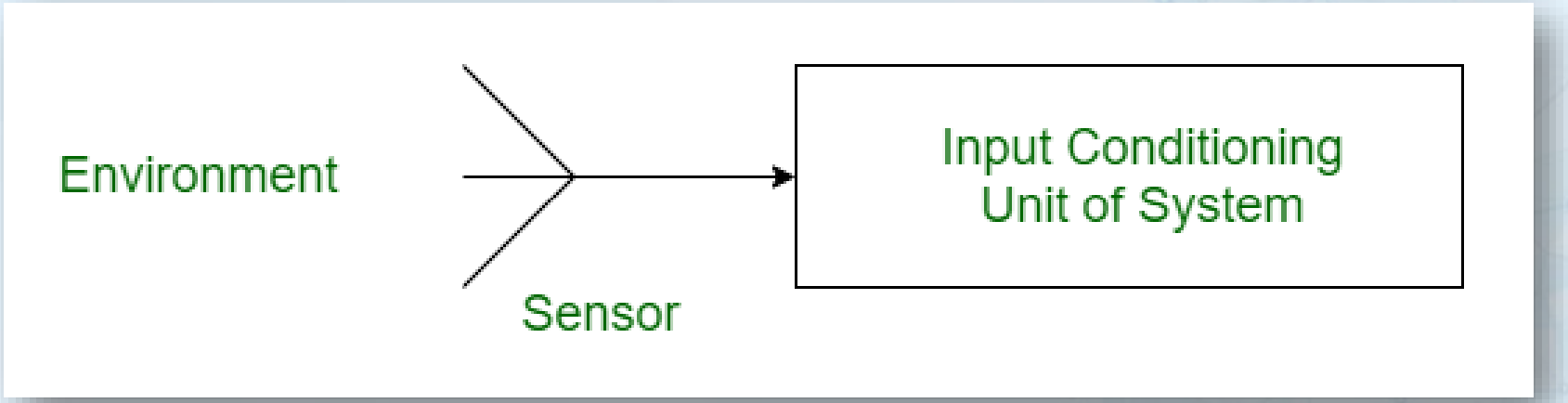
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Sensors & Actuators

- Sensor is a device used for the conversion of physical events into the electrical signals.
- This is a hardware device that takes the input from environment and gives to the system by converting it.
- For example, a thermometer takes the temperature as physical characteristic and then converts it into electrical signals for the system.
- Ex:
  - Temperature sensors, Light sensor, Pressure sensor , Motion Sensor.

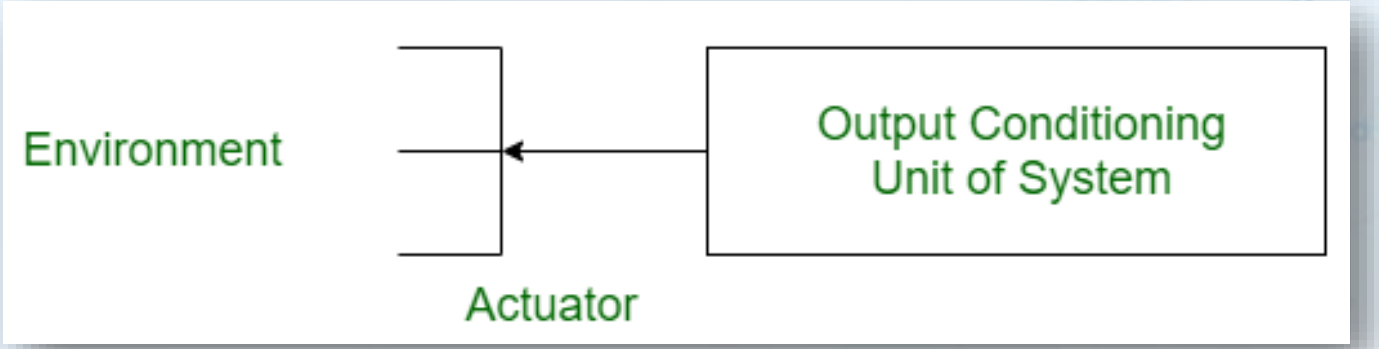
Sensors & Actuators

- Sensor



Sensors & Actuators

- Actuator is a device that converts the electrical signals into the physical events.
- It takes the input from the system and gives output to the environment.
- For example, motors and heaters are some of the commonly used actuators.



Conclusion

- IC
- MPU
- MCU
- CPU
- GPU
- DSP
- MCU vs SoC
- ECU

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History of Embedded System

- The first recognized embedded system is the Apollo Guidance Computer (AGC) developed by MIT lab.
- It used 5000 ICs.
- The UI of AGC is known DSKY(display/keyboard) which resembles a calculator type keypad with array of numerals.
- In the year 1971 Intel introduced the world's first microprocessor chip called the 4004, was designed for use in business calculators. It was produced by the Japanese company Busicom.



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Classification of Embedded System

- The classification of embedded system is based on following criteria:
  - On generation
  - On complexity & performance
  - On deterministic behavior
  - On triggering

Classification of Embedded System

1. Embedded Systems - Classification based on Generation:

- First Generation: The early embedded systems built around 8-bit microprocessors and 4-bit microcontrollers



EX. Digital Telephone Keypads etc.

- Second Generation: Embedded Systems built around 16-bit microprocessors and 8 or 16-bit microcontrollers.

EX. SCADA, Data Acquisition Systems etc.

Classification of Embedded System

1. Embedded Systems - Classification based on Generation:

- Third Generation: Embedded Systems built around high performance 16/32-bit Microprocessors/controllers, Application Specific Instruction set processors like Digital Signal Processors (DSPs), and Application Specific Integrated Circuits (ASICs). The instruction set is complex and powerful.
- EX. Robotics, industrial process control, networking etc.

Classification of Embedded System

1. Embedded Systems - Classification based on Generation:

- Fourth Generation: Embedded Systems built around System on Chips (SoC's), Reconfigurable processors and multicore processors. It brings high performance, tight integration and miniaturization into the embedded device market
- EX Smart phone devices, MIDs etc.

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Classification of Embedded System

2. Embedded Systems - Classification based on Complexity & Performance

1. Small-scale:

- Simple in application need.
- Performance not time-critical.
- Built around low performance & low cost.

Example: an electronic toy



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Classification of Embedded System

2. Embedded Systems - Classification based on Complexity & Performance

2. Medium-scale:

- Slightly complex in hardware & firmware requirement.
- Built around medium performance & low cost.
- Usually contain operating system.
- Examples: Industrial machines.

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Classification of Embedded System

2. Embedded Systems - Classification based on Complexity & Performance

3. Large-scale:

- Highly complex hardware & firmware.
- Response is time-critical.

Ex: Mission critical applications.

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Classification of Embedded System

2. Embedded Systems - Classification based on deterministic behavior

It is applicable for Real Time systems.

• These are classified in to two types:

1. *Soft Real time Systems:* Missing a deadline may not be critical and can be

tolerated to a certain degree.

2. *Hard Real time systems:* Missing a program/task execution time deadline can have catastrophic consequences (financial, human loss of life, etc.)

Classification of Embedded System

2. Embedded Systems - Classification based on Triggering:

These are classified into two types

1. Event Triggered: Activities within the system (e.g., task run-times) are dynamic and depend upon occurrence of different events .

2. Time triggered: Activities within the system follow a statically computed schedule (i.e., they are allocated time slots during which they can take place) and thus by nature are predictable.

Application of Embedded System

- The application areas and the products in the embedded domain are countless.
  1. *Consumer Electronics:* Camcorders, Cameras.





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Application of Embedded System

- The application areas and the products in the embedded domain are countless.

  2. *Household appliances:* Washing machine, Refrigerator.





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Application of Embedded System

- The application areas and the products in the embedded domain are countless.

  3. *Automotive industry:* Anti-lock braking system(ABS), engine control.

  ABS: is a vehicle safety feature that prevents wheels from locking up during heavy braking, allowing the driver to maintain steering control and stability.

  It works by using wheel speed sensors and an electronic control unit (ECU) to rapidly apply and release the brakes, cycling the pressure to keep the wheels rotating. You'll know ABS is working when you feel a pulsing or hear a grinding noise in the brake pedal.

Application of Embedded System

- The application areas and the products in the embedded domain are countless.

  4. *Home automation & security systems:* Air conditioners, sprinklers, fire alarms.





Application of Embedded System

- The application areas and the products in the embedded domain are countless.

  5. *Telecom:* Cellular phones, telephone switches.



Application of Embedded System

- The application areas and the products in the embedded domain are countless.

6. *Computer peripherals:* Printers, scanners.
7. *Computer networking systems:* Network routers and switches.
8. *Healthcare:* Different Kinds of Scanners, EEG, ECG machines.
9. *Banking & Retail:* Automatic teller machines, point of sales.
10. *Card Readers:* Barcode, smart card readers.



Purpose of Embedded Systems:

- Each Embedded Systems is designed to serve the purpose of any one or a combination of the following tasks:
  - Data Collection/Storage/Representation
  - Data Communication
  - Data (Signal) Processing
  - Monitoring
  - Control
  - Application Specific User Interface

Purpose of Embedded Systems:



1. Data Collection/Storage/Representation:-
   - Performs acquisition of data from the external world.
   - The collected data can be either analog or digital.
   - Data collection is usually done for storage, analysis, manipulation and transmission.
   - The collected data may be stored directly in the system or may be transmitted to some other systems, or it may be processed by the system, or it may be deleted instantly after giving a meaningful representation.

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Purpose of Embedded Systems:



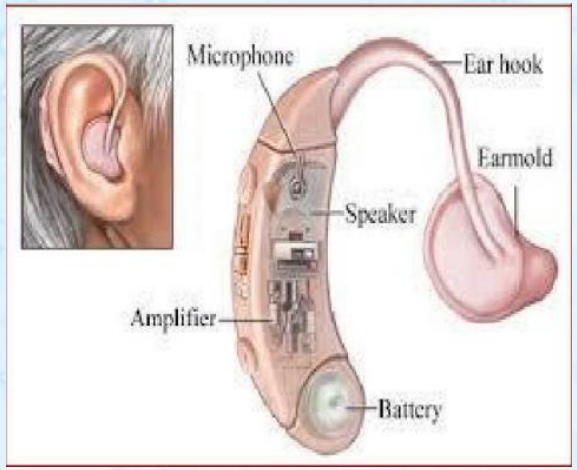
2. Data Communication:-

- Embedded Data communication systems are deployed in applications *ranging from* complex satellite communication systems to simple home networking systems.
- Embedded Data communication systems are dedicated for data communication.
- The data communication can happen through a wired interface (like Ethernet, etc.) or wireless interface (like Wi-Fi, Bluetooth, etc.).
- Network hubs, Routers, switches, Modems etc. are typical examples for dedicated data transmission embedded systems.

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Purpose of Embedded Systems:

3. Data (Signal) Processing:-
   - Embedded systems with Signal processing functionalities are employed in applications demanding signal processing **like** Speech coding, synthesis, audio video codec, transmission applications etc.
   - Computationally intensive systems.
   - Employs Digital Signal Processors (DSPs).



Purpose of Embedded Systems:



4. Monitoring:-

- Embedded systems coming under this category are specifically designed for monitoring purpose.
- They are used for determining the state of some variables using input sensors.
- They cannot impose control over variables (Just Monitoring).
- Electro Cardiogram (ECG) machine for monitoring the heart beat of a patient is a typical example for this.
- Measuring instruments like Digital CRO, Digital Multi meter, Logic Analyzer etc. used in Control & Instrumentation applications are also examples of embedded systems for monitoring purpose.

Purpose of Embedded Systems:



5. Control:-

- Embedded systems with control functionalities are used for imposing control over some variables according to the changes in input variables.
- Embedded system with control functionality contains both sensors and actuators.
- Sensors are connected to the input port for capturing the changes in environmental variable or measuring variable.
- The actuators connected to the output port are controlled according to the changes in input variable to put an impact on the controlling variable to bring the controlled variable to the specified range.

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Purpose of Embedded Systems:



5. Control:-
   - Air conditioner for controlling room temperature is a typical example for embedded system with **Control** functionality.
   - Air conditioner contains a room temperature sensing element (sensor) which may be a thermistor and a handheld unit for setting up (feeding) the desired temperature.
   - The air compressor unit acts as the actuator. The compressor is controlled according to the current room temperature and the desired temperature set by the end user.

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Purpose of Embedded Systems:

6. Application Specific User Interface:-
   - Embedded systems which are designed for a specific application.
   - Contains Application Specific User interface (rather than general standard UI) like key board, Display units etc.
   - Aimed at a specific target group of users.
   - Mobile handsets, Control units in industrial applications etc. are example.



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THANK YOU FOR WATCHING

QUESTIONS?

